

DBCLS BioHackathon 2026 report: A knowledge base for the Marchantia genome database MarpolBase

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Introduction

The liverwort *Marchantia polymorpha* (Bowman et al., 2017) has drawn attention as a new-generation model plant in developmental and evolutionary biology, and a high-quality reference genome has been determined for it (Tanizawa et al., 2025). For bioinformatics the species is attractive for a further reason: the community maintains an agreed gene nomenclature and gene identifier system, and researchers use that system when they describe genes in their papers. A gene mentioned in the literature can therefore be resolved to an identifier with far less ambiguity than in most species, which is an advantage both for extracting information from papers and for integrating data across sources.

Our goal at the DBCLS BioHackathon 2026 was to turn **MarpolBase** (Tanizawa et al., 2025), the genome database for this species, into a knowledge base: to publish its gene, expression and literature data as RDF, to expose them through a SPARQL endpoint, and to make them reachable from AI agents through a Model Context Protocol (MCP) server.

The question behind this was practical. *M. polymorpha* is not indexed by the major cross-reference hubs, so it is not obvious that a species-specific resource can be linked to the wider set of life-science databases at all. Much of what follows is therefore about *where* the join actually happens.

Table 1 lists what was produced.

Table 1: Resources and tools developed or extended during BioHackathon 2026.

Name	Description and URL
MarpolBase RDF	RDF distribution of the MarpolBase gene, expression and literature data. https://marchantia.info/rdf/
MarpolBase SPARQL endpoint	Public endpoint, registered with TogoMCP and RDF Portal. https://marchantia.info/sparql
MarpolBase MCP server	14 tools giving agents typed access to genes, expression, co-expression and literature. https://marchantia.info/mcp

MarpolBase RDF and MCP server

What was published

The gene, expression and literature data served by MarpolBase were converted to RDF and released. The SPARQL endpoint was registered with both TogoMCP and RDF Portal, so MarpolBase can now be reached from the cross-database query tooling maintained in Japan. On top of the endpoint we built an MCP server with 14 tools, which lets an LLM agent query the resource without writing SPARQL by hand.

The tools fall into five groups: gene lookup and search, expression profiles and condition comparison, co-expression networks, curated gene–literature links, and controlled-vocabulary resolution. A `run_sparql` tool and a `describe_schema` tool are also provided so that an agent can fall back to raw SPARQL when the typed tools are not enough.

Use cases

We exercised the server with two analyses driven entirely from natural-language prompts.

In the first (Figure 1), the prompt “*tell me the conditions in which MpBNB is highly expressed*” was answered by retrieving the mean TPM of MpBNB across all 161 conditions in MarpolBase, together with the condition attributes (tissue, accession, stage, sex, mutant, treatment, infection). The gene reaches 27.8 TPM in the antheridium and stays near zero in thallus, gemma and spore, which the agent could report because the expression values carry condition metadata rather than bare sample identifiers.

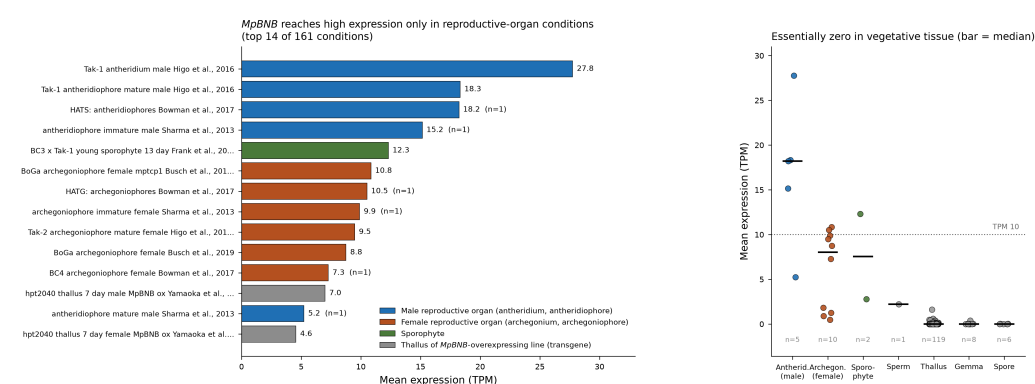
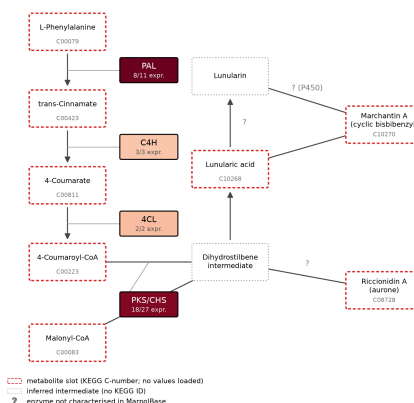


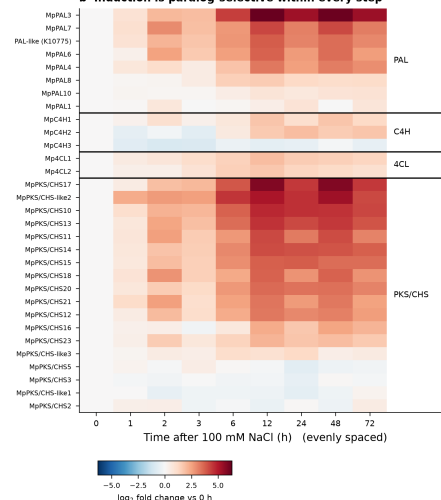
Figure 1: Expression of *MpBNB* across the 161 conditions in MarpolBase, retrieved through the MCP server from a single natural-language prompt. Left: top 14 conditions by mean TPM, coloured by organ type. Right: the same values grouped by tissue.

In the second (Figure 2), we followed the bisbibenzyl biosynthesis route under salt stress. Expression values came from MarpolBase; the pathway context — compound identifiers, reactions and enzyme references — came from external resources reached through TogoMCP. The committing PKS/CHS step and the shared phenylpropanoid entry are both salt-induced, and the induction is paralogue-selective within every step.

a Bisbibenzyl route: the committing PKS/CHS step is salt-induced, like the shared phenylpropanoid entry



b Induction is paralog-selective within every step



Panel a: enzyme boxes coloured by the strongest induction among that step's expressed paralogs; 'N exp' = paralogs above the expression floor / all paralogs annotated in MarpolBase. Panel b: rows are individual genes (log2 fold change vs 0 h), grouped by step and sorted by 12 h response. 31 of 43 genes shown; 12 omitted — 11 below the 1 TPM floor (a fold change from a -0.01 TPM baseline is not interpretable) and MgPAL5 (MgPAL5 [4170], whose profile MarpolBase returned an error for and was therefore not assessed, not treated as absent). Steps from lunularic acid onward have no characterized enzymes in MarpolBase and are left uncoloured. Reaction topology is a literature schematic, NOT from KEGG — lunularic acid (C00088), marchantin A (C00070) and riccionidin A (C00728) exist in KEGG COMPOUND but are placed on no KEGG pathway map. Expression: Tai-1 thallus 7 d, Wu et al., 2021 (n_runs 3-7 per time point).

Figure 2: The bisbibenzyl route under salt stress. (a) Pathway schematic with enzyme steps coloured by the strongest induction among the expressed paralogs of that step. (b) log2 fold change of individual genes over a 100 mM NaCl time course, grouped by step.

What MarpolBase and TogoMCP each contribute

Working through these two analyses made the division of labour between a species-specific resource and a cross-database hub concrete (Table 2).

Only MarpolBase can supply measured expression across its 161 conditions, the co-expression network (PCC / HRR / MR), DESeq2 differential expression between two conditions, gene-literature links curated with evidence and role, and controlled-vocabulary resolution with PECO and PO identifiers. It also carries its own KEGG Orthology assignments — `mpo:hasKEGG` for 6,563 genes. This last point mattered in practice: *M. polymorpha* is absent from KEGG GENES, and the enzyme nodes in Figure 2 could be coloured **only** because MarpolBase assigns KO numbers itself.

Only TogoMCP can convert identifiers across 119 databases (TogoID) (Ikeda et al., 2022), search the species-agnostic references (UniProt, PDB, Reactome, Rhea, ChEMBL, MeSH), reach the NCBI E-utilities for SRA, GEO and PubMed, and dispatch SPARQL to a choice of endpoints.

Table 2: Where the two resources overlap, and how the overlap differs in kind.

Capability	MarpolBase	TogoMCP
SPARQL	One own graph (GRAPH must be given explicitly)	A choice of endpoints
Literature	<i>Marchantia</i> papers linked to genes, with evidence and role	Generic PubMed / MeSH search
Compounds	Not covered	ChEBI / Rhea / PubChem identifier conversion
Gene expression	Measured values for 161 conditions	Not held

The two do not meet at the gene identifier

The most useful negative result of the project: **TogoID holds no *M. polymorpha* gene dataset, and MarpolBase's `get_gene` does not return UniProt accessions.** There is therefore no identifier-conversion path that carries Mp3g23300 directly into the TogoMCP side. We also confirmed that pfam → uniprot does not exist as a TogoID route (404), while rhea/chebi, chebi/pubchem_compound, uniprot/rhea, uniprot/pdb and go/uniprot do (in both directions).

The two resources meet instead at the level of **shared vocabularies**:

1. KO numbers — the junction that made the two-layer map in Figure 2 possible.
2. GO and Pfam — assigned by MarpolBase, dereferenceable on the TogoMCP side.
3. Compound identifiers (ChEBI / PubChem) — the junction for metabolome data.

Going through UniProt instead means searching by protein name and organism, which for paralog families such as MpPAL1–10 is many-to-many and therefore unreliable.

The practical rule we arrived at: use MarpolBase for measured values, conditions, co-expression and curated literature in *M. polymorpha*; use TogoMCP to carry those results outward to structures, reactions, compounds, orthologs in other species and sequence archives; and bridge the two with KO, GO and compound identifiers rather than with gene identifiers.

Discussion

Publishing MarpolBase as RDF made it possible to link a species-specific plant genome resource to external resources, and registering the endpoint with TogoMCP and RDF Portal contributed a plant genome resource to the shared Japanese infrastructure. The most transferable lesson is the one in Table 2 and the section that follows it: for a resource covering a species that the major cross-reference hubs do not index, the integration point is not the gene identifier but the shared vocabularies the resource chooses to assign — in our case, above all, its own KO assignments.

The MCP server also changed how the resource is used in practice. Both analyses in this report were driven from a natural-language prompt rather than from hand-written SPARQL, and the value of the typed tools was less in saving keystrokes than in carrying the condition metadata along with the numbers, so that the agent could report *where* a gene is expressed and not only *how much*.

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Author contributions

Yasuhiro Tanizawa: Conceptualization, Software, Writing – original draft; Takatomo Fujisawa: Software, Writing – review & editing.

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